

Revision Notes for Class 12 History Chapter 10 Computer Networks

We live in a highly connected world where computers, devices, and people are part of large communication networks. A computer network is formed when two or more computers or digital devices are linked together, allowing them to share information and resources.

Common examples include not just computers, but also printers, mobile phones, and storage devices, all communicating through wired or wireless means.

Networks make it possible for users to send emails, browse websites, place audio or video calls, and even access a shared printer from different computers.

Devices that are part of a network and can send, create, or receive data are called nodes. These nodes can be computers, smartphones, switches, routers, or other hardware like printers.

Each network is made up of networking devices such as switches, routers, and modems. These devices help in connecting and managing different computers at homes, schools, offices, and across continents. For secure and efficient communication, data sent over a network is broken into packets, allowing for fast transfer and reliable delivery.

Evolution of Computer Networks

The roots of computer networking started in the 1960s with the ARPANET research project in the US. ARPANET aimed to connect different academic and research centers so that scientists could effectively collaborate. Over the years, this small, specialized network evolved into the modern Internet.

- 1969: First successful communication between UCLA and SRI via ARPANET.
- 1971: Email introduced; @ symbol used for addresses.

- 1983: TCP/IP protocol adopted, forming the base of the Internet.
- 1990: The invention of HTML and URLs led to the birth of the World Wide Web.
- 1997: Introduction of Wi-Fi standards, enabling wireless connectivity.

Types of Networks

Computer networks can be classified according to area coverage and data speed. The main types are:

- **PAN (Personal Area Network):** Covers a small area (10 metres), like connecting a phone to a laptop using Bluetooth.
- **LAN (Local Area Network):** Connects computers within a room, building, or campus. Offers high speed and secure data transfer.
- **MAN (Metropolitan Area Network):** Spans a city using cable or wireless. Commonly seen in cable television or broadband networks.
- **WAN (Wide Area Network):** Connects large geographical regions or the world. The Internet is the biggest WAN.

Networking Devices and Their Functions

To set up and maintain networks, several devices are required. Each plays a distinct role in ensuring smooth communication:

- **Modem (MODulator DEModulator):** Converts digital data from computers to analog signals for transmission over telephone lines or vice versa.
- **Network Interface Card (NIC):** Allows a device to connect to a wired Ethernet network and is identified by a unique MAC address.
- **RJ45:** A connector with eight pins used to plug Ethernet cables into devices.

- **Repeater:** Amplifies weak signals over long distances, restoring them to original strength.
- **Hub:** Broadcasts data to all connected devices; can cause data collision if many devices transmit at once.
- **Switch:** Sends data only to the intended device on the network, improving efficiency and reducing collisions.
- **Router:** Connects multiple networks, such as local networks to the Internet, and decides the best path for data.
- **Gateway:** Acts as a bridge between organizational networks and external networks, often equipped with security features.

Network Topologies

Network topology is the pattern or arrangement in which computers and peripherals are connected. Different topologies are suitable for different needs:

Mesh Topology: Every device is linked with every other device. Offers reliability, but is difficult to install due to the large number of cables.

Ring Topology: Each device is connected to two others, forming a circle. Data travels in one direction and passes through each device.

Bus Topology: All devices are connected to a single backbone cable. Easy to implement but has a single point of failure.

Star Topology: All devices connect to a central hub or switch. It is efficient and easy to troubleshoot, but the central device is a point of failure.

Tree or Hybrid Topology: Combination of star, ring, and bus. Used for complex networks, especially in organizations with WANs and many LANs.

Identifying Devices: MAC and IP Address

Each networked device must have a unique way to be identified. This is achieved through two main addressing systems: MAC and IP addresses.

MAC Address: Media Access Control address is a 48-bit number hard-coded into the NIC hardware; it is permanent and unique to each device. Example: 00-14-22-01-23-45

IP Address: Internet Protocol address is assigned by software protocols and can change as a device moves between networks. IPv4 uses 32 bits (e.g., 192.168.0.178); IPv6 uses 128 bits (e.g., 2001:CDBA:0000:0000:0000:0000:3257:9652).

Internet, World Wide Web, and Internet of Things (IoT)

The Internet is the largest and most far-reaching network, connecting devices such as computers, smartphones, printers, cameras, and more. Local ISPs provide Internet access, which is then joined to the global network backbone.

The World Wide Web (WWW) is a collection of web pages and resources that we access using Internet browsers. It operates using three main technologies: HTML for content creation, URLs for addresses, and HTTP/HTTPS for communication.

- **HTML (HyperText Markup Language):** Designs standard web pages.
- **URL (Uniform Resource Locator):** Tells where to find resources like websites or documents.
- **HTTP (HyperText Transfer Protocol):** The set of rules for browsing pages.

Today, IoT (Internet of Things) is growing quickly. Devices such as smart TVs, air conditioners, vehicles, and security cameras now connect to the Internet, broadening the range of networked devices beyond traditional computers and phones.

Domain Name System (DNS)

Remembering complex numeric IP addresses for websites is challenging. DNS (Domain Name System) solves this problem by assigning readable names to each web resource and linking them to their numeric IP addresses.

When a user enters a website address, the DNS server translates it into its corresponding IP address so data can be retrieved from the right location. The process, called domain name resolution, is done automatically in the background every time you access a web page.

DNS servers are organized in a hierarchy, with 13 root servers at the top managed by the Internet Assigned Numbers Authority (IANA). This system ensures that websites and services can be accessed reliably by billions of users worldwide.

Summary of Key Points

- A computer network is an interconnection of two or more computers or digital devices for sharing data and resources.
- The Internet is the largest WAN, with billions of devices connected globally.
- Common types of networks include PAN, LAN, MAN, and WAN, each serving different area sizes.
- Key hardware devices include modems, switches, routers, hubs, repeaters, gateways, and NICs.
- Network topology defines how devices are linked, with options like mesh, star, ring, bus, and hybrid structures.
- Devices are identified using MAC addresses (hardware-based) and IP addresses (software-assigned).
- The World Wide Web uses HTML, URL, and HTTP, making information sharing possible through web browsers.

- DNS makes Internet navigation easy by linking domain names to numeric IP addresses.

